



Exception Handling

Practical Skills Module 4: Topic 5

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Topic Code	PM-04-PS05
Topic	Handle exceptions

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0501 Catch and handle exceptions

Applied Knowledge

- AK0501 Concept, definition and functions of intermediate Python programming principles

Internal Assessment Criteria

- IAC0501 Expected results with Python exception handling are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-04



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario:

A retail business needs to process customer orders from a file. However, the file might not exist, or it could contain invalid data. To ensure smooth program

execution, you need to **catch and handle exceptions** such as `FileNotFoundError` or `ValueError`.

1. Create a Python file called `process_orders.py`

2. Create a text file for testing called `customer_orders.txt` and add the following content into it

```
ORD001, John Doe, 100.50
ORD002, Jane Smith, 200.75
ORD003, Alice Johnson, abc
```

Save the file in the **same directory** as your Python script.

Step 1: Set up your file reading code

- In the python file you created open `customer_orders.txt` for reading
 - Read each line of the file
 - Split each line into `order_id`, `customer_name`, and `amount`

Step 2: Handle `FileNotFoundError`

- Implement exception handling techniques to
 - Print a helpful message when the textfile is not found

Step 3: Handle `ValueError`

Implement exception handling techniques to:

- Detect when an amount value from the file cannot be processed as a number
 - Ensure the program does not crash and continues processing the next line

- Display a helpful message when this type of data issue occurs

Step 4: Ensure graceful completion

- Use a `finally` block to close the file if it was opened
- Add a message at the end to confirm the program completed successfully

Example Output

```
Processing orders...
Order ORD001 by John Doe for R100.5
Order ORD002 by Jane Smith for R200.75
Invalid amount in line: ORD003, Alice Johnson, abc
Done processing.
```

Your facilitator will complete the following evaluation checklist:

Check that the following is accomplished:

Item	Checked (Yes=5 No=0)	Comment: where did you find the evidence?
PA0501 Catch and handle exceptions		
Name of member		
Signature		
Date		
Total		/5



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
